

PhysioVision

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Session 2021-2025

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June, 2025

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Chapter 1

Introduction

Arthritis is a medical condition characterized by joint pain, stiffness, swelling, and inflammation, leading to a reduced range of motion and fatigue. In more severe cases, it can result in joint deformities and permanent disability. According to the World Health Organization (WHO), arthritis affects approximately 528 million people globally, marking a 113 percent increase since 1990 [2]. Around 60% of arthritis patients undergo physiotherapy to alleviate pain, with 70% of them experiencing moderate pain relief and 30% reporting significant improvement [1].

1.1 Problem Statement

In Pakistan, access to physiotherapy treatment is limited due to high costs and overburdened healthcare systems. Many individuals in need of treatment are unable to receive consistent care, leading to prolonged suffering and decreased quality of life due to the prolonged pain becoming taxing on their productivity and mental health.

1.2 Scope

This project is an AI-driven application that provides personalized physiotherapy routines and nutrition plans for arthritis recovery. By combining a chatbot with a computer vision system, it delivers customized diet guidance and exercise suggestions tailored to each user's needs.

The exercises suggested by the chatbot, personalized for each individual, are then performed under the guidance of a real-time correction model. This computer vision model monitors user angles, corrects exercise postures, and provides real-time feedback through a multilingual audio bot in English and Urdu, as well as visual on-screen cues. It gen-

erates detailed session reports, which, along with user feedback, are fed back into the chatbot to adjust future routines and diet plans based on the user's progress.

1.3 Modules

1.3.1 Module 1 - Vision Model

The vision model monitors and analyzes user exercise postures in real time, providing corrections and feedback to ensure proper form. It uses computer vision to track angles and movement, delivering guidance through multilingual audio and visual cues while generating detailed session reports for ongoing adjustments

1. Real-Time Posture Monitoring
2. Live Posture Correction
3. Audio-Bot
4. Visual Cues
5. Multi-language support for audio-bot and visual cues.
6. Session Reports
7. User Feedback

1.3.2 Module 2 - Chat Bot

This module recommends personalized physiotherapy routines and nutrition plans based on user input and progress. It integrates with user feedback and session reports to continually refine and adapt exercise and diet recommendations for optimal arthritis recovery.

1. Nutrition Plan
2. Exercise Recommendation
3. Dynamic Recommendations

1.3.3 Module 3- Client Web Application

The Client Web Application offers an intuitive interface for users to access and manage their personalized physiotherapy and nutrition plans. Integrated with the AI chat bot and vision model, it ensures tailored and adaptive support for optimal arthritis recovery.

1. User-Friendly Interface

1.4 User Classes and Characteristics

User Class	Description
Patients	Individuals suffering from arthritis who want to efficiently manage their treatment through personalized physiotherapy routines and nutrition plans from the comfort of their homes.
Physiotherapist	Healthcare professionals who provide physiotherapy services and will use the app to track patient progress, recommend exercises, and manage treatment plans more effectively.
Caregiver	Family members or friends assisting patients in managing their arthritis treatment. They will use the app to monitor progress and ensure adherence to exercise and nutrition plans.
AI/ML Researcher	Researchers in the fields of medicine or AI/ML interested in analyzing the algorithms used in the app to assess their effectiveness and explore potential improvements.

Table 1.1: User Classes and Their Characteristics

Chapter 2

Project Requirements

This chapter explores the core elements that define how the system operates. We'll dive into use cases that detail how our user will interact with the system, followed by an overview of the functional and non-functional requirements that ensure the system meets both user needs and technical standards and finally domain model for a conceptual view of the system.

2.1 Use-cases

2.1.1 Use-Case Diagram

The use case diagram provides a high-level overview of the key interactions between the user and the system.

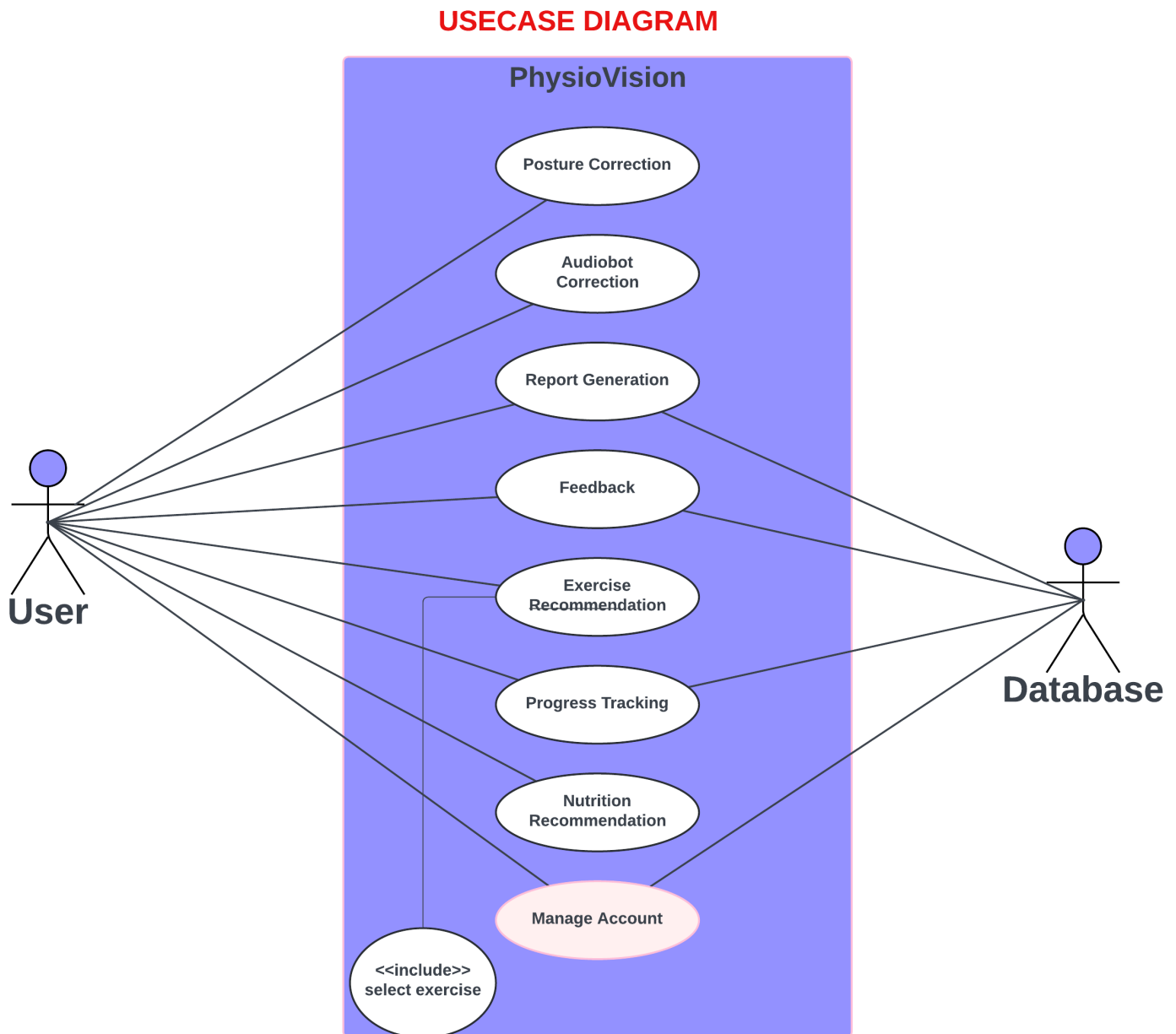


Figure 2.1: Use-Case Diagram of User Interactions with the System

2.1.2 Exercise Recommendation via Chatbot

Use Case Name	Exercise Recommendation via Chatbot
Scope	Personalized Physiotherapy Routine Generation
Level	User Goal
Primary Actor	User (Arthritis Patient)
Stakeholders	User: Wants to receive a personalized exercise routine based on their arthritis condition. AI Physiotherapist (Chatbot): Responsible for recommending exercises. Healthcare Providers: Ensure the exercise recommendations support patient health. System Admin: Ensures the chatbot operates smoothly and follows health guidelines.
Preconditions	1. The user has logged into the application. 2. The user has provided necessary health-related data (e.g., age, arthritis severity, pain level, mobility restrictions). 3. The chatbot is active and ready to provide recommendations. 4. Past session reports (if any) are available to refine the recommendations.
Post Conditions	Success: The chatbot provides a personalized exercise routine, which the user performs under the guidance of the real-time correction model.
	Failure: An error message appears if the chatbot fails to generate an exercise plan or if critical data is missing.
Main Success Scenario	1. The user opens the chatbot and requests an exercise plan. 2. The chatbot retrieves the user's health profile, mobility data, and past session reports. 3. The AI generates a personalized exercise routine suited to the user's arthritis condition. 4. The chatbot provides detailed instructions for each exercise. 5. The user performs the exercises under the supervision of the real-time correction model.

Alternatives	Incomplete Health Data: If the system detects missing or incomplete health data, the chatbot prompts the user to fill in the required information before generating recommendations. User Requests Adjustments: If the user feels pain during an exercise, the chatbot adjusts the routine by suggesting easier alternatives. The chatbot also provides additional advice based on user feedback during the session. Chatbot Unavailable: If the chatbot is down, the system suggests accessing previously saved exercise routines until service is restored.
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Table 2.1: Use Case 1: Exercise Plan

2.1.3 Nutrition Recommendation via Chatbot

Use Case Name	Nutrition Recommendation via Chatbot
Scope	Personalized Nutrition Plan Generation
Level	User Goal
Primary Actor	User (Arthritis Patient)
Stakeholders	User: Seeks a personalized nutrition plan that supports arthritis recovery. AI Nutritionist (Chatbot): Provides tailored dietary recommendations. Healthcare Providers: Ensure the dietary plan is appropriate for the patient's condition. System Admin: Maintains the chatbot's smooth operation and adherence to health guidelines.
Preconditions	1. The user is logged into the application. 2. The user has provided health and dietary information (e.g., arthritis severity, dietary restrictions, preferences, allergies). 3. The chatbot is available to interact and provide recommendations. 4. Past feedback or session reports (if applicable) are available to refine the recommendations.
Post Conditions	Success: The chatbot delivers a personalized nutrition plan, which the user can follow to support arthritis management. The system logs user feedback on the nutrition plan for future adjustments. Failure: An error message is shown if the chatbot cannot generate a nutrition plan due to missing data or technical issues.

Main Success Scenario	1. The user initiates a conversation with the chatbot and requests a personalized nutrition plan. 2. The chatbot retrieves the user's health data, including any dietary restrictions or preferences. 3. The AI system generates a custom diet plan that focuses on anti-inflammatory foods, appropriate portion sizes, and essential nutrients to support arthritis recovery. 4. The chatbot provides a detailed breakdown of recommended meals, including breakfast, lunch, dinner, and snacks. 5. The chatbot offers practical meal suggestions, along with tips on food preparation and ingredient alternatives. 6. The system logs the user's feedback on the plan, allowing future recommendations to be refined based on preferences and health progress.
Alternatives	Incomplete Dietary Information: If the user has not provided complete dietary data (e.g., allergies, preferences), the chatbot asks for additional information before generating a plan. User Requests Specific Adjustments: If the user prefers certain foods or has specific dietary needs, the chatbot modifies the plan to accommodate those preferences without compromising nutritional goals. Chatbot Unavailable: If the chatbot is unavailable due to technical issues, the system allows the user to access pre-saved nutrition plans or meal ideas.

Table 2.2: Use Case 2: Nutrition Plan

2.1.4 Real-Time Exercise Posture Correction

Use Case Name	Real-Time Exercise Posture Correction
Scope	Exercise Posture Monitoring and Correction
Level	User Goal
Primary Actor	User (Arthritis Patient)
Stakeholders	User: Seeks immediate feedback on exercise posture to avoid injury and maximize effectiveness. AI Physiotherapist (Computer Vision Model): Responsible for monitoring and correcting the user's posture in real-time. Healthcare Providers: Want to ensure exercises are performed safely and correctly. System Admin: Ensures the computer vision model functions correctly.
Preconditions	1. The user has initiated an exercise routine via the system. 2. The camera and computer vision model are operational. 3. The user's posture is within the model's field of view.
Post Conditions	Success: The computer vision model successfully monitors the user's posture and provides real-time corrections. The system logs the user's performance for future analysis.
	Failure: The system alerts the user if posture cannot be monitored due to camera issues or incorrect placement.
Main Success Scenario	1. The user begins performing exercises as recommended by the system. 2. The computer vision model monitors the user's body angles and movement in real-time. 3. If incorrect posture or joint movement is detected, the system highlights the issue visually on-screen. 4. The system suggests immediate corrections (e.g., "Straighten your back," "Bend your knees slightly"). 5. The user adjusts their posture according to the feedback, and the system continues monitoring until the session is complete.

Alternatives	Camera Malfunction or Poor Visibility: If the camera is malfunctioning or the user is out of frame, the system alerts the user to adjust the camera or their position. Posture Not Corrected: If the user doesn't correct their posture after multiple prompts, the system suggests stopping the exercise to avoid injury.
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Table 2.3: Use Case 3: Posture Correction

2.1.5 Audiobot Exercise Posture Correction

Use Case Name	Audiobot Exercise Posture Correction
Scope	Multilingual Audio Feedback for Exercise Correction
Level	User Goal
Primary Actor	User (Arthritis Patient)
Stakeholders	User: Receives verbal cues in real-time to adjust posture while exercising. Audiobot: Provides correction instructions in the user's preferred language (English/Urdu). Healthcare Providers: Ensure patients perform exercises safely through guided corrections. System Admin: Maintains the functionality of the audiobot.
Preconditions	1. The user is performing exercises recommended by the system. 2. The computer vision model detects incorrect posture. 3. The audiobot is active and ready to provide feedback in the selected language.
Post Conditions	Success: The audiobot provides real-time audio feedback to correct exercise posture in the user's chosen language (English/Urdu). The system logs feedback sessions for future adjustments.
	Failure: An error message appears if the audiobot is unable to provide feedback due to technical issues.
Main Success Scenario	1. The user begins their exercise routine, and the computer vision model monitors posture. 2. When the model detects incorrect posture, the audiobot provides real-time correction via verbal cues (e.g., "Straighten your back" in English or "Kamar seedhi karain" in Urdu). 3. The user adjusts their posture according to the audio instructions while continuing the exercise. 4. The audiobot continues providing corrections as needed, ensuring safe and effective exercise performance.

Alternatives	Audiobot Unavailable: If the audiobot is unavailable, the system defaults to visual on-screen feedback only. Incorrect Posture Not Corrected After Audio Cues: If the user does not correct their posture after multiple audio prompts, the system suggests stopping the exercise to prevent injury.
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Table 2.4: Use Case 4: Audiobot

2.1.6 Exercise Session Report Generation

Use Case Name	Exercise Session Report Generation
Scope	Exercise Session Data Analysis and Report Creation
Level	User Goal
Primary Actor	User (Arthritis Patient)
Stakeholders	User: Receives detailed reports on exercise performance and posture correction to track progress and guide future routines. AI Physiotherapist (System): Responsible for generating the report based on exercise session data. Healthcare Providers: May review reports to monitor the user's progress and make adjustments to the therapy if necessary. System Admin: Ensures accurate and timely report generation.
Preconditions	1. The user has completed an exercise session. 2. Data from the completed session is logged by the system (exercise performance, posture corrections, feedback). 3. The system is operational and able to access past session data, if available.
Post Conditions	Success: A comprehensive report is generated, summarizing the user's exercise performance, posture corrections, and improvements.
	Failure: An error occurs if session data is incomplete or if the system is unable to generate the report.

Main Success Scenario	1. The user completes a scheduled exercise session. 2. The system collects relevant data, including: (a) Number of repetitions and sets completed for each exercise. (b) Any posture corrections made during the session (based on real-time feedback). (c) Duration of the session. (d) Any feedback from the user regarding discomfort or pain during the exercises. 3. The AI system analyzes the data and generates a detailed report. 4. The report includes: (a) Exercise Performance: Number of completed repetitions, sets, and overall time spent on each exercise. (b) Posture Corrections: A summary of real-time posture corrections and improvements during the session. (c) User Feedback: Details on how the user felt during the session, including any pain or discomfort experienced. (d) Progress: Comparisons with previous sessions, highlighting improvements or areas for further attention. 5. The report is displayed to the user, with visual charts showing progress over time (e.g., improvement in posture, increase in repetitions). 6. The report is stored in the system and used by the AI to adjust future exercise routines based on the user's progress.
Alternatives	Incomplete Session Data: If the session data is incomplete, the system generates a partial report and notifies the user of missing data (e.g., skipped exercises, incomplete feedback). User Requests Detailed Analysis: If the user requests more detailed feedback, the system generates an in-depth report comparing performance over multiple sessions, showing trends in posture correction, repetition increase, and more. System Unavailable: If the system is unavailable or data is corrupted, the user is notified that the report could not be generated and advised to try again later.

Table 2.5: Use Case 5: Session Report

2.1.7 User Feedback Collection and Integration

Use Case Name	User Feedback Collection and Integration
Scope	Collecting and Using User Feedback to Adjust Future Exercise Routines
Level	User Goal
Primary Actor	User (Arthritis Patient)
Stakeholders	User: Provides feedback on exercise sessions to influence future recommendations. AI Physiotherapist (System): Collects and uses feedback to adjust future exercise routines. Healthcare Providers: May use feedback data to monitor the user's experience and adjust treatment plans if needed. System Admin: Ensures the feedback system operates smoothly and integrates properly with the AI.
Preconditions	1. The user has completed an exercise session. 2. The system is ready to collect feedback immediately after the session.
Post Conditions	Success: The user's feedback is successfully collected, stored, and integrated into future exercise routine recommendations.
	Failure: Feedback collection fails due to system issues, or the user does not submit feedback.

Main Success Scenario	1. After completing an exercise session, the system prompts the user to provide feedback. 2. The user submits feedback on various aspects, including: (a) Exercise Difficulty: Whether the exercises were too hard, too easy, or appropriately challenging. (b) Pain or Discomfort: Whether any exercises caused pain or discomfort, and if so, in which joints or areas. (c) Session Satisfaction: General satisfaction with the session and how it helped their recovery. (d) Posture Correction: How helpful the real-time feedback was during the exercises. 3. The system collects the feedback and logs it for analysis. 4. The AI system uses the feedback to adjust future exercise routines, focusing on: (a) Reducing or increasing the intensity based on the user's reported experience. (b) Replacing exercises that caused discomfort with safer alternatives. (c) Adjusting posture correction prompts based on user preference and feedback. 5. The feedback is stored alongside the user's session data for future reference and progress tracking. 6. The system confirms with the user that their feedback has been successfully received and will be integrated into the next session.
Alternatives	User Skips Feedback: If the user chooses not to provide feedback, the system logs the session data but does not adjust the routine. A reminder is shown during the next session. Incomplete Feedback: If the user only partially submits feedback, the system analyzes the available information and adjusts recommendations accordingly, while prompting for missing details later. System Unavailable: If the feedback system is down, the user is notified and prompted to provide feedback after the issue is resolved, with an option to provide offline feedback for future use.

Table 2.6: Use Case 6: User Feedback

2.1.8 Progress Tracking for Exercise Performance

Use Case Name	Progress Tracking for Exercise Performance
Scope	Monitoring and Analyzing User Progress Over Time
Level	User Goal
Primary Actor	User (Arthritis Patient)
Stakeholders	User: Seeks to monitor progress in exercise performance and overall recovery. AI Physiotherapist (System): Responsible for tracking and analyzing user performance data. Healthcare Providers: May review progress reports to adjust treatment plans based on user progress. System Admin: Ensures that progress tracking functionalities work correctly and data is securely stored.
Preconditions	1. The user has completed multiple exercise sessions. 2. The system has collected and stored performance data from these sessions.
Post Conditions	Success: The system successfully tracks and displays the user's progress over time, helping to inform future exercise routines and motivate the user.
	Failure: An error occurs if data is incomplete or if the system is unable to retrieve performance history.

Main Success Scenario	1. The user logs into the application and navigates to the progress tracking section. 2. The system retrieves historical exercise performance data, including: (a) Number of repetitions and sets completed over time. (b) Posture correction data and improvements. (c) User feedback collected after each session. (d) Any reported pain or discomfort associated with specific exercises. 3. The system analyzes the data and generates a visual representation of progress, such as: (a) Graphs showing improvements in exercise performance (e.g., increased repetitions or improved form). (b) Charts highlighting trends in reported discomfort or pain levels associated with specific exercises. (c) Summaries of session satisfaction ratings over time. 4. The user reviews their progress, noting areas of improvement and any exercises that may require modification based on pain levels. 5. The system may suggest future exercises or adjustments based on progress, encouraging users to set new goals for their rehabilitation. 6. The user can export or share progress reports with health-care providers if needed.
Alternatives	Insufficient Data for Tracking: If there is insufficient data to provide a meaningful analysis, the system informs the user and encourages them to continue logging sessions for better tracking. User Requests Detailed Progress Analysis: If the user wants more in-depth analysis, the system can generate a comprehensive report comparing performance across multiple sessions or specific time frames. System Unavailable: If the progress tracking feature is down, the user is notified and given the option to receive updates once the feature is restored.

Table 2.7: Use Case 7: Progress Tracking

2.1.9 Account Management for User Profiles

Use Case Name	Account Management for User Profiles
Scope	Managing User Accounts and Personal Information
Level	User Goal
Primary Actor	User (Arthritis Patient)
Stakeholders	User: Manages personal information, preferences, and account settings. System: Ensures the security and accuracy of user account data. Healthcare Providers: May access account information to better understand user preferences and treatment needs. System Admin: Responsible for maintaining the integrity and security of user accounts.
Preconditions	1. The user has registered and created an account in the application. 2. The user has successfully logged into the system.
Post Conditions	Success: The user successfully updates their account information, preferences, and settings. Any changes made are saved securely and reflected in future sessions.
	Failure: An error occurs if the system is unable to save changes or if invalid information is entered.

Main Success Scenario	1. The user logs into their account and navigates to the account management section. 2. The user can view and edit the following account details: (a) Personal Information: Name, email address, phone number, etc. (b) Preferences: Language settings (English/Urdu), notification preferences, etc. (c) Health Information: Any relevant health details, exercise preferences, and goals. (d) Password Management: Options to change or reset the password. 3. The user makes the desired changes and submits the updates. 4. The system validates the information entered (e.g., checks for valid email format, password strength). 5. Upon successful validation, the system saves the changes and updates the user's profile. 6. The user receives a confirmation message indicating that their account details have been successfully updated. 7. The user can log out, and the system retains all updated information for the next login.
Alternatives	Invalid Information Entered: If the user enters invalid data (e.g., an incorrectly formatted email), the system displays an error message and prompts the user to correct it. Password Reset Requested: If the user forgets their password, they can initiate a password reset process that involves verifying their identity through email. System Unavailable: If the account management feature is down, the user is notified that they cannot make changes at that time and advised to try again later.

Table 2.8: Use Case 8: Manage Account

2.2 Functional Requirements

Listed below are the functional requirements of our system, organized by modules.

2.2.1 Module 1 - Vision Model

Following are the requirements for module 1:

1. The system shall generate a detailed report after each session, summarizing user performance, posture errors, and corrections made.
2. The system shall track user progress over time by comparing session reports.
3. The system shall provide visual graphs or data summaries on a weekly or monthly basis to show improvements or regressions.
4. The system shall use a computer vision model to detect key body joints and positions in real-time during exercises.
5. The system shall analyze body positions and angles to detect deviations from correct posture and identify common mistakes (e.g., slouching, incorrect limb angles).
6. The system shall provide real-time feedback for incorrect posture via audio or on-screen text.
7. The system shall offer specific guidance to correct posture errors (e.g., "Raise your arms higher" or "Straighten your back").

2.2.2 Module 2 - Chat Bot

Following are the requirements for module 2:

1. The system shall allow users to input personal data (e.g., height, weight, arthritis severity).
2. The system shall collect feedback from users on exercise difficulty, pain level, and progress.
3. The system shall track exercise completion using data from the computer vision model.
4. The system shall generate personalized physiotherapy routines based on user inputs (e.g., arthritis severity, physical ability).

5. The system shall suggest tailored nutrition plans to support arthritis management based on user inputs.
6. The system shall generate weekly or monthly reports on user progress, including pain improvement, exercise completion, and dietary adherence.

2.3 Non-Functional Requirements

2.3.1 Multilingual Support for Audio-Bot and Visual Cues

1. The audio-bot and on-screen feedback must be available in multiple languages (including English and Urdu).
2. Users should be able to select their preferred language for both audio and visual guidance.

2.3.2 Performance

1. The system must provide real-time posture analysis and feedback within a maximum delay of 200 milliseconds to ensure a seamless user experience.
2. It should support continuous operation during exercise sessions without noticeable lag or delays in corrections.

2.3.3 Compatibility

1. The system must be compatible with a variety of devices, including smartphones, tablets, and computers, across different operating systems (e.g., iOS, Android, Windows).
2. It should work seamlessly across different screen sizes and resolutions.

2.3.4 Scalability

1. The system should be scalable to accommodate future features, such as new exercises, languages, or additional types of therapy for other conditions.

2.3.5 Real-Time Interaction

1. The chatbot must respond to user queries and inputs within a minimal latency period for a smooth user experience.

2.3.6 User-Friendly Interface

1. The chatbot's user interface must be intuitive, with easy navigation for people of all ages, especially arthritis patients who may have limited mobility.

2.3.7 Accessibility

1. The chatbot must be designed with accessibility in mind, supporting voice commands and text input for users with physical impairments.

2.3.8 Error Handling

1. The chatbot should gracefully handle errors (e.g., unrecognized commands or incomplete data input) by providing appropriate feedback to users and attempting to correct issues.

2.3.9 User Authentication

1. Authentication must be secure, following industry standards for data protection (e.g., HTTPS, encryption).

2.3.10 User-Friendly Interface for Web Application

1. The web application must allow users to easily access and manage their personalized physiotherapy and nutrition plans.
2. It must integrate with the AI chatbot to provide real-time feedback and guidance based on user input.

2.3.11 Availability

1. The Client Web Application must achieve high uptime, ensuring continuous access to physiotherapy and nutrition plans.

2.3.12 Concurrent User Access

1. The Client Web Application must support concurrent access by multiple users without performance degradation, ensuring a seamless experience for all us

2.4 Domain Model

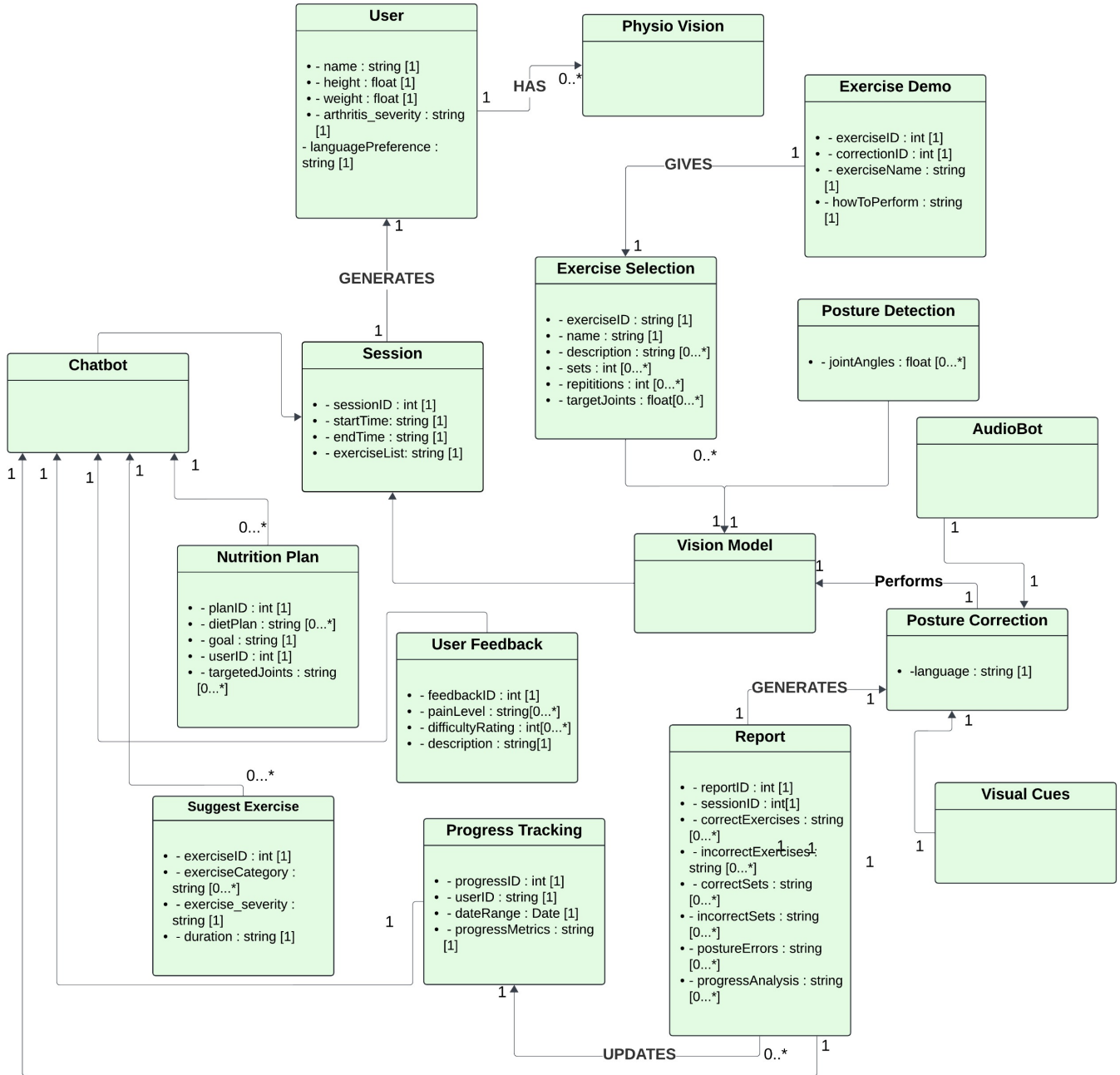


Figure 2.2: Domain Model PhysioVision

Chapter 3

System Overview

This chapter will analyze the architectural design of the project, focusing on key modules and their interactions. It will explore how the system transitions between different states and how data flows through various components, providing a concise overview of the design's functionality and context. By breaking down the architecture and illustrating relationships between modules, this chapter aims to clarify the overall structure and operational dynamics of the system.

3.1 Architectural Design

3.1.1 Box and Line Diagram

The Box and Line Diagram visually represents the system's architecture, highlighting key modules and their interconnections. It features a User Interface that encompasses a Chatbot Interface for personalized exercise and nutrition suggestions, and a Vision Model Interface for posture detection and correction. The Database module serves as a centralized repository for user profiles and progress tracking, supporting all system components. Overall, the diagram illustrates how each module collaborates to provide a comprehensive support platform for users' health journeys.

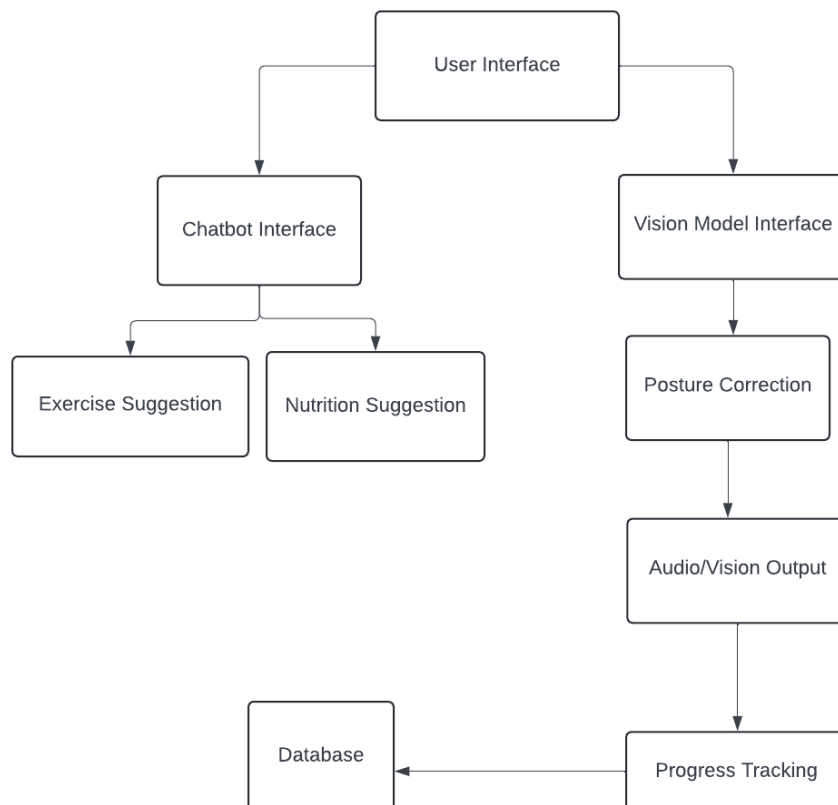


Figure 3.1: Box and Line Diagram of the System Architecture

3.1.2 Layered Architecture Diagram

The architecture diagram illustrates a multi-layered structure comprising three primary layers: Presentation, Business Logic, and Database. The Presentation Layer includes user interfaces such as the Chatbot Interface and Vision Model Interface, enabling users to interact with the system. This layer sends requests to the Business Logic Layer, which processes these requests through modules like Nutrition Recommendation and Posture Detection Model, facilitating tasks such as report generation and posture correction. The Database Layer serves as the backend, storing essential data such as exercise records, user progress and user profiles, handling read/write operations from the Business Logic Layer. This architectural model supports a clear separation of concerns, allowing for efficient communication between user interfaces and backend processes, enhancing system maintainability and scalability.

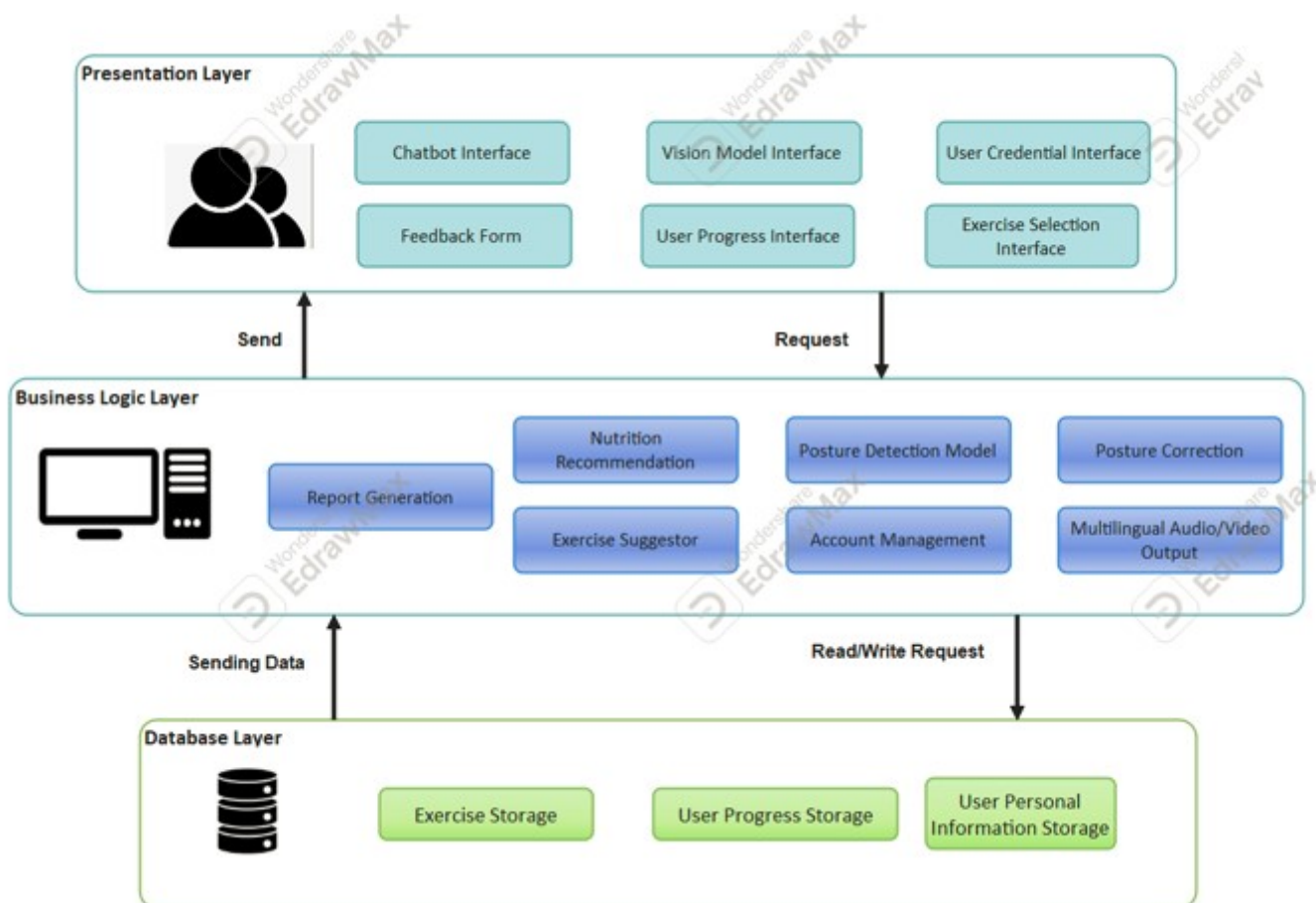


Figure 3.2: Layered Architecture Diagram

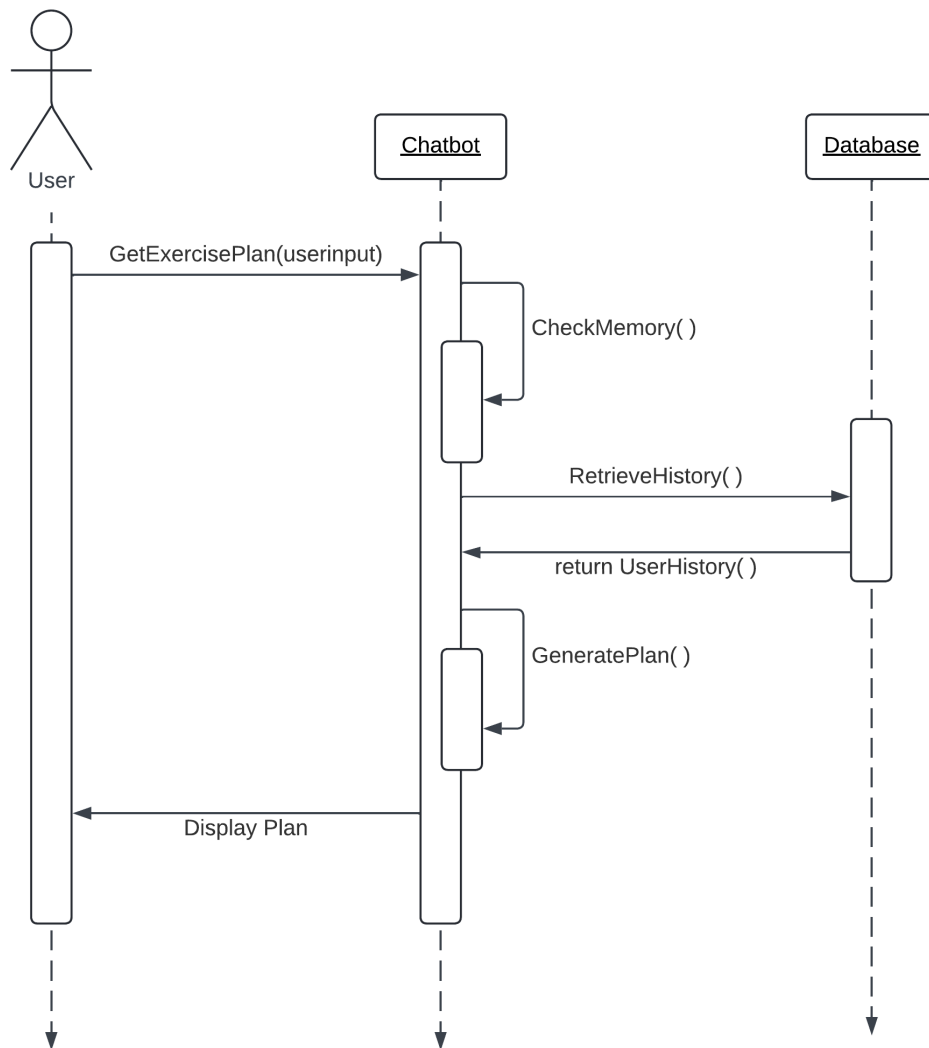
3.2 Design Models

The following design models outline the project's architecture, detailing the interactions and relationships between system components. They serve as a visual representation of how different modules collaborate to deliver a seamless user experience. These models provide a comprehensive framework for understanding the system's functionality and operational flow.

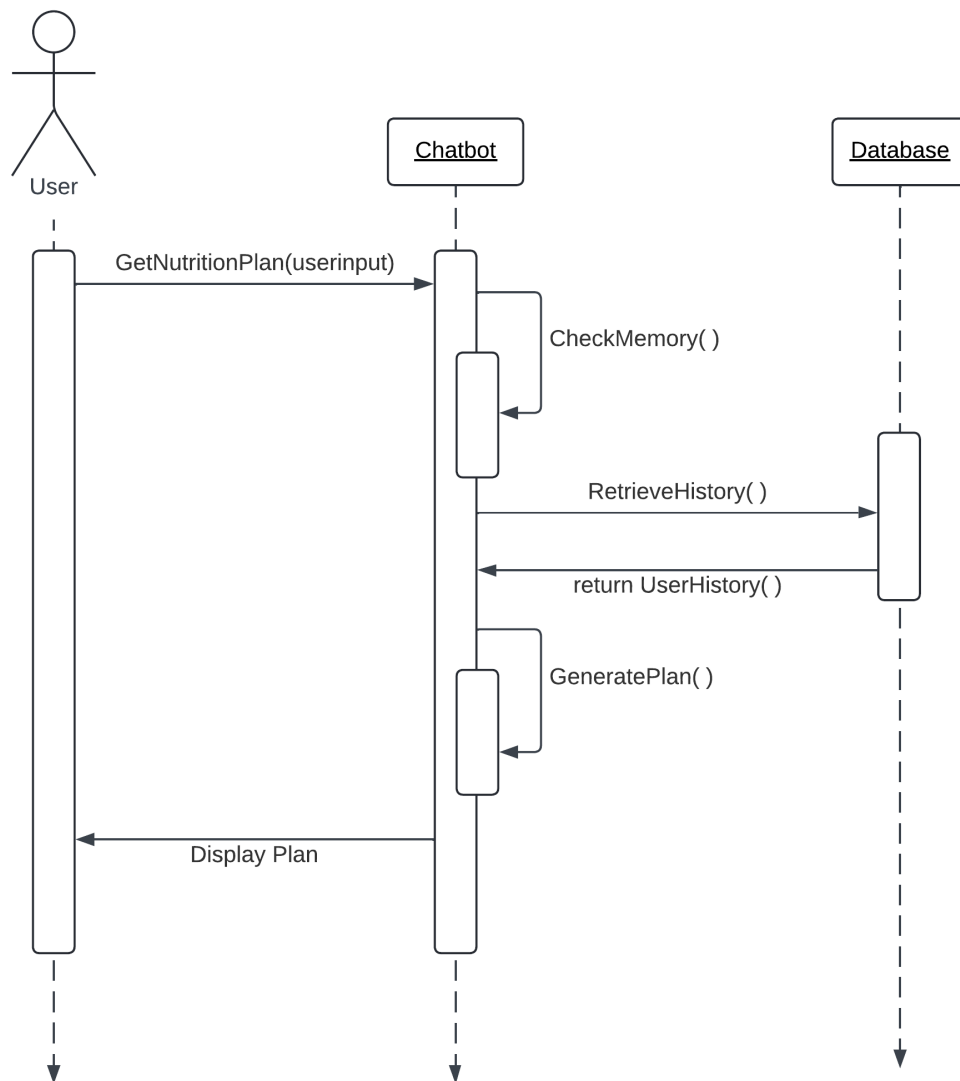
3.2.1 Sequence Diagrams

The following Sequence Diagrams are being used to visualize the flow of messages or interactions between different objects or components in the system for each particular usecase.

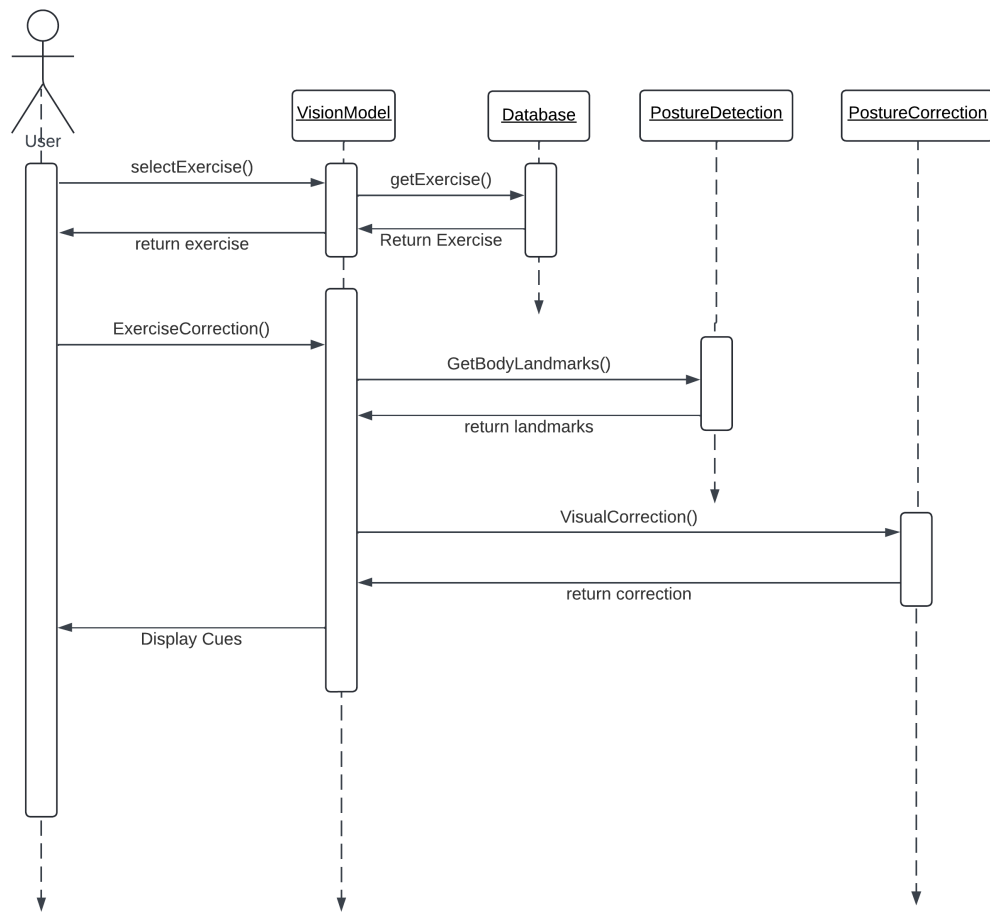
3.2.1.1 Exercise Recommendation via Chatbot



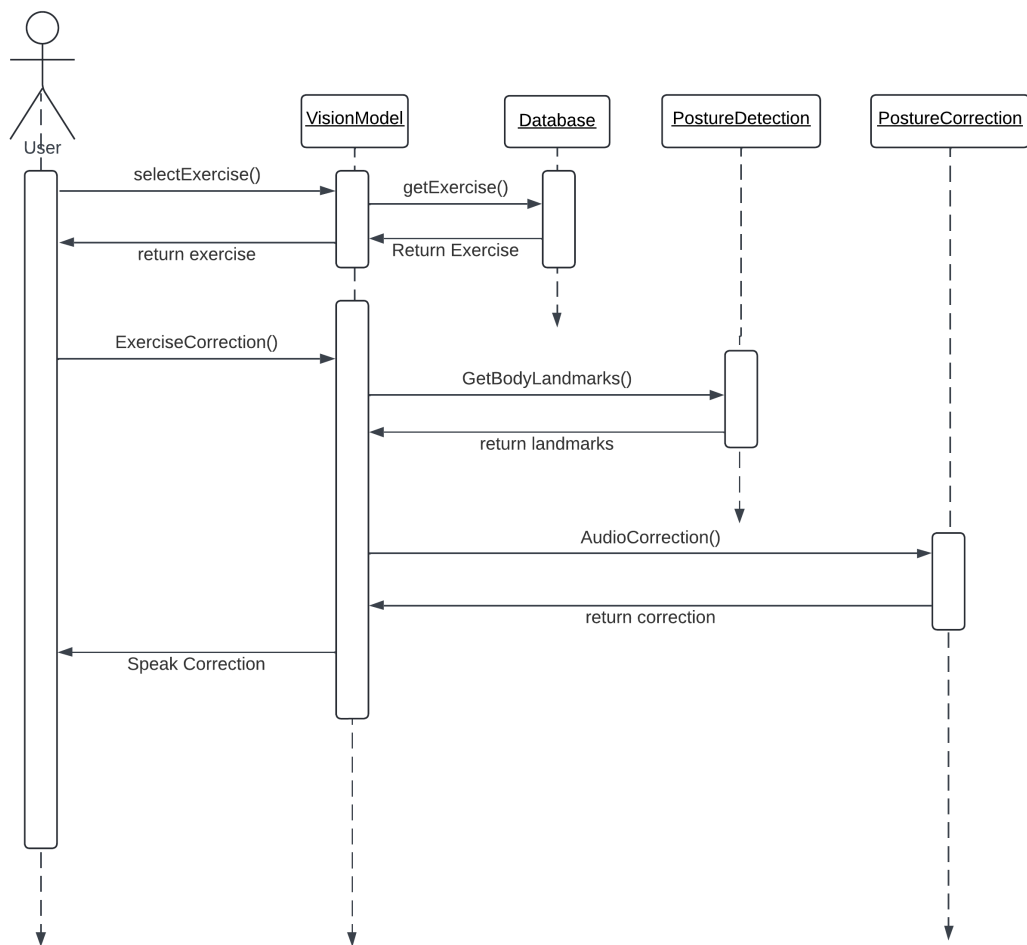
3.2.1.2 Nutrition Recommendation via Chatbot



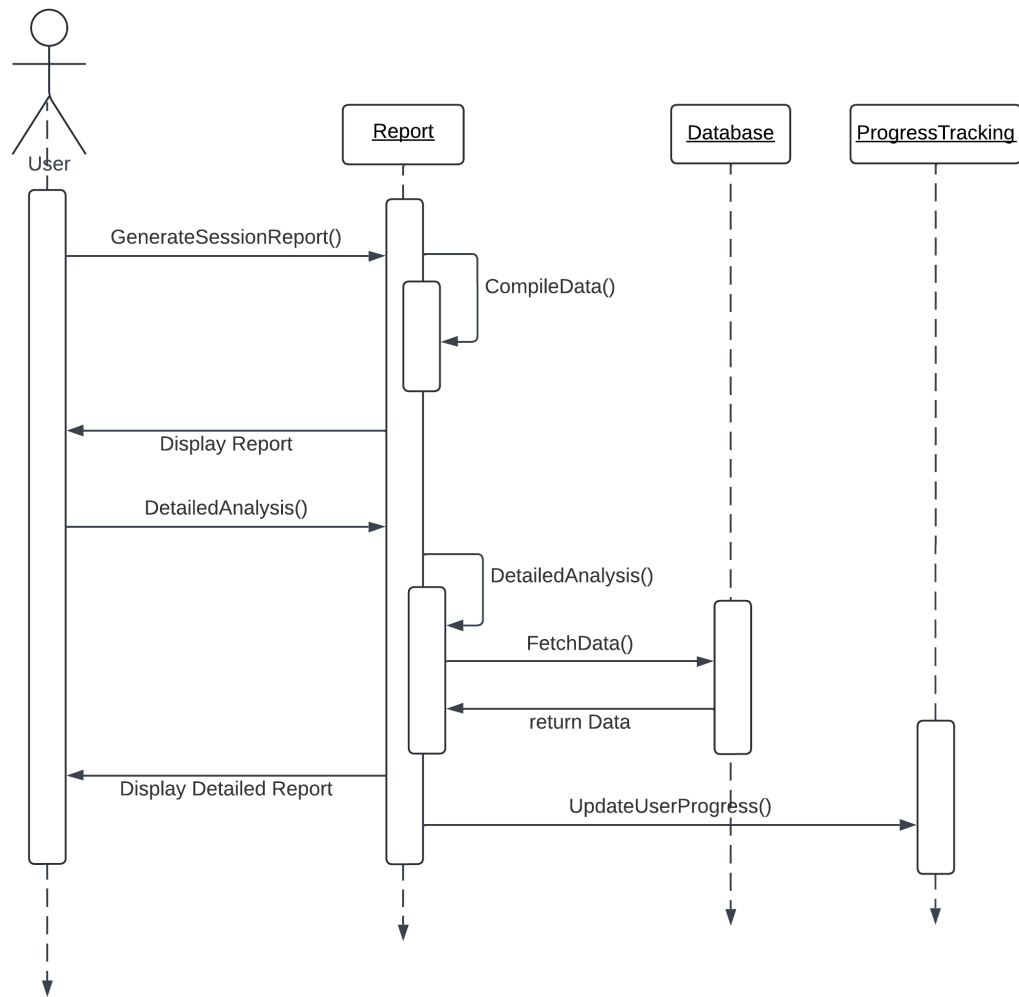
3.2.1.3 Real-Time Exercise Posture Correction



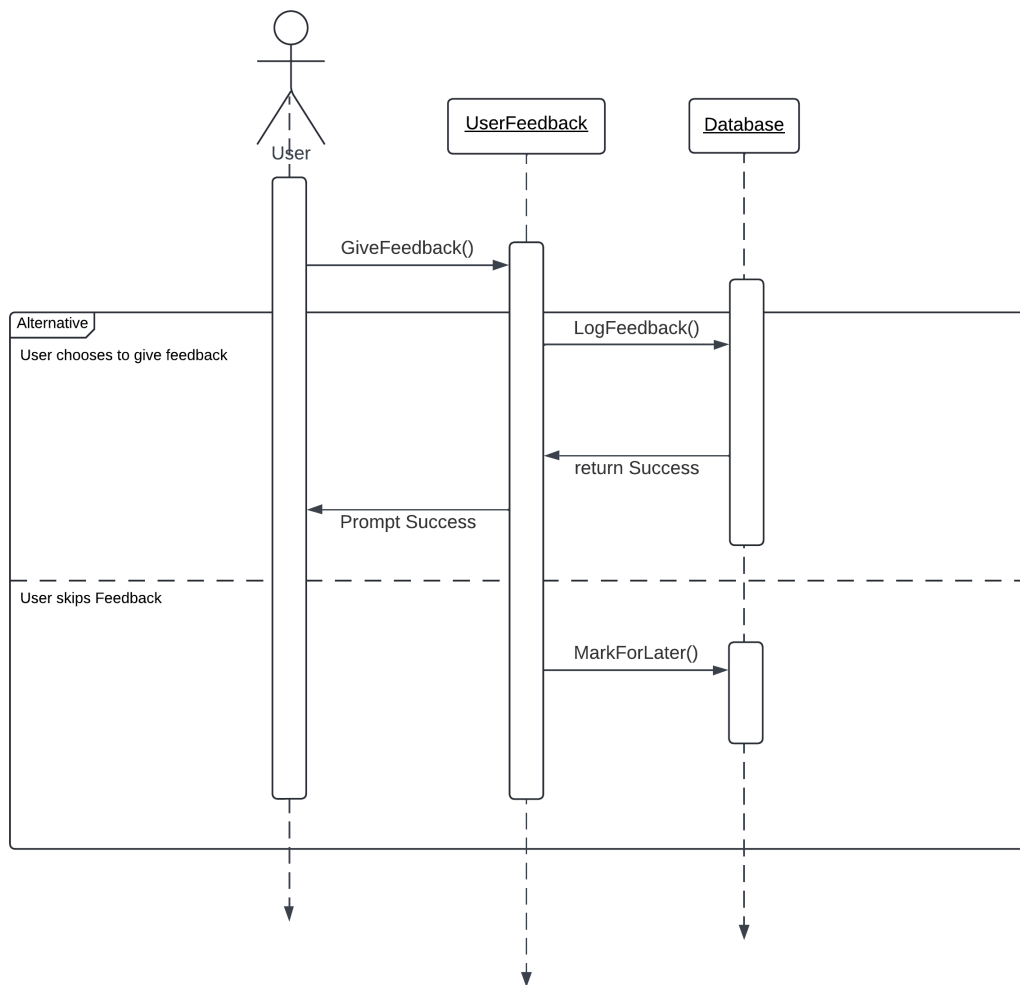
3.2.1.4 Audiobot Exercise Posture Correction



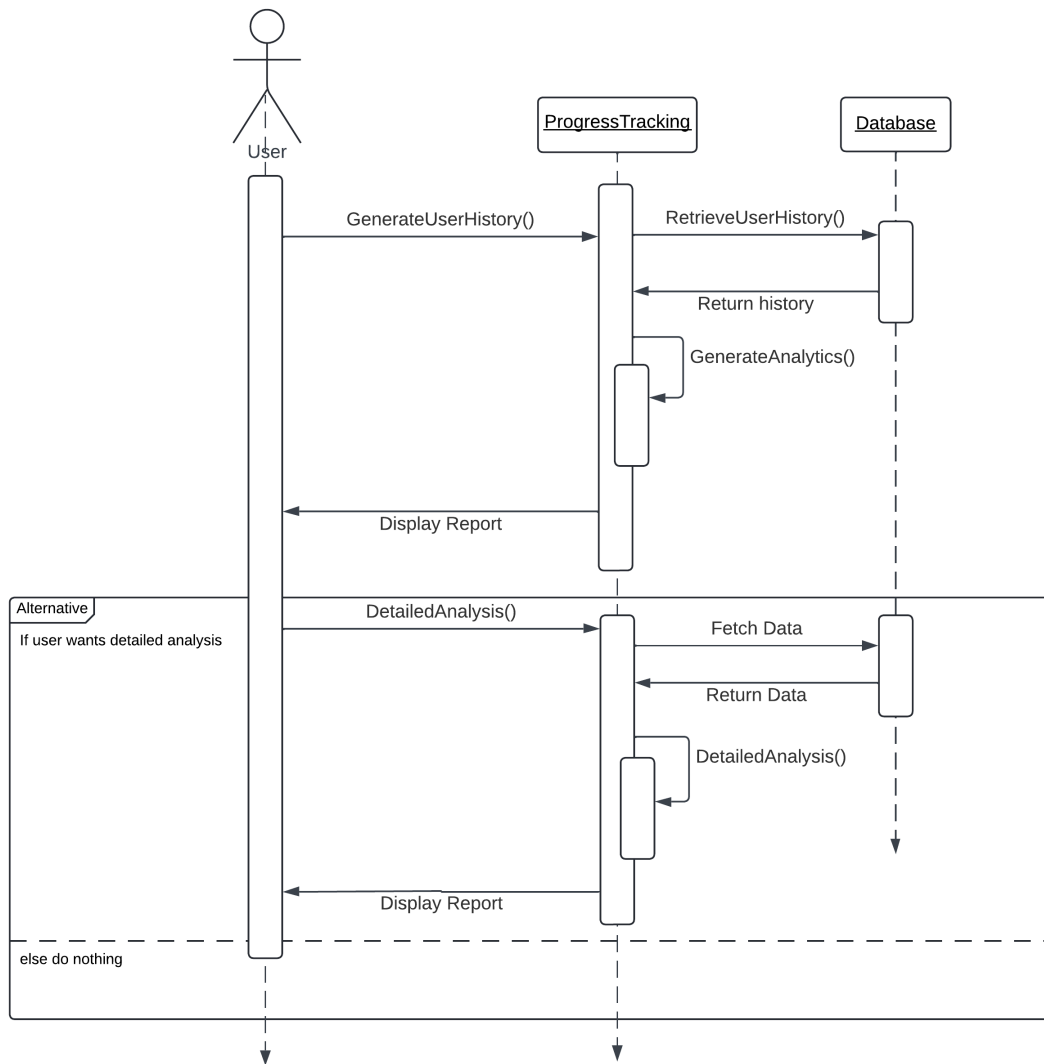
3.2.1.5 Exercise Session Report Generation



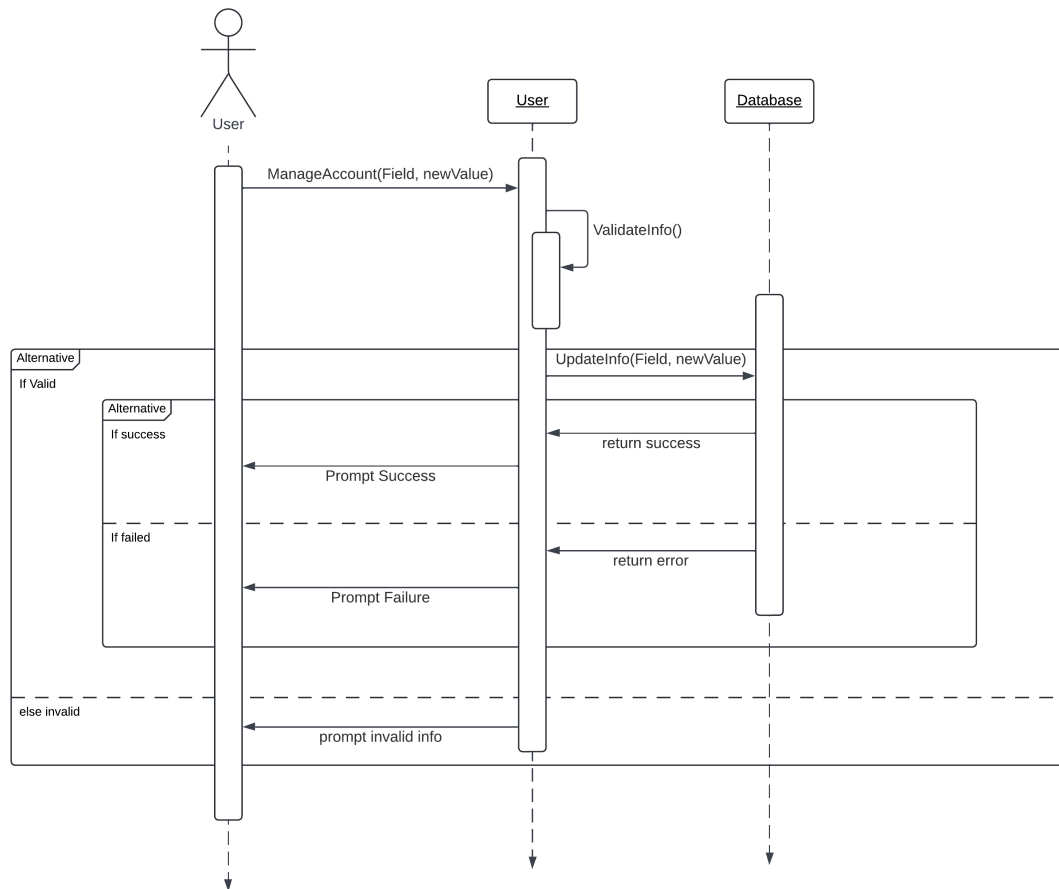
3.2.1.6 User Feedback Collection and Integration



3.2.1.7 Progress Tracking for Exercise Performance



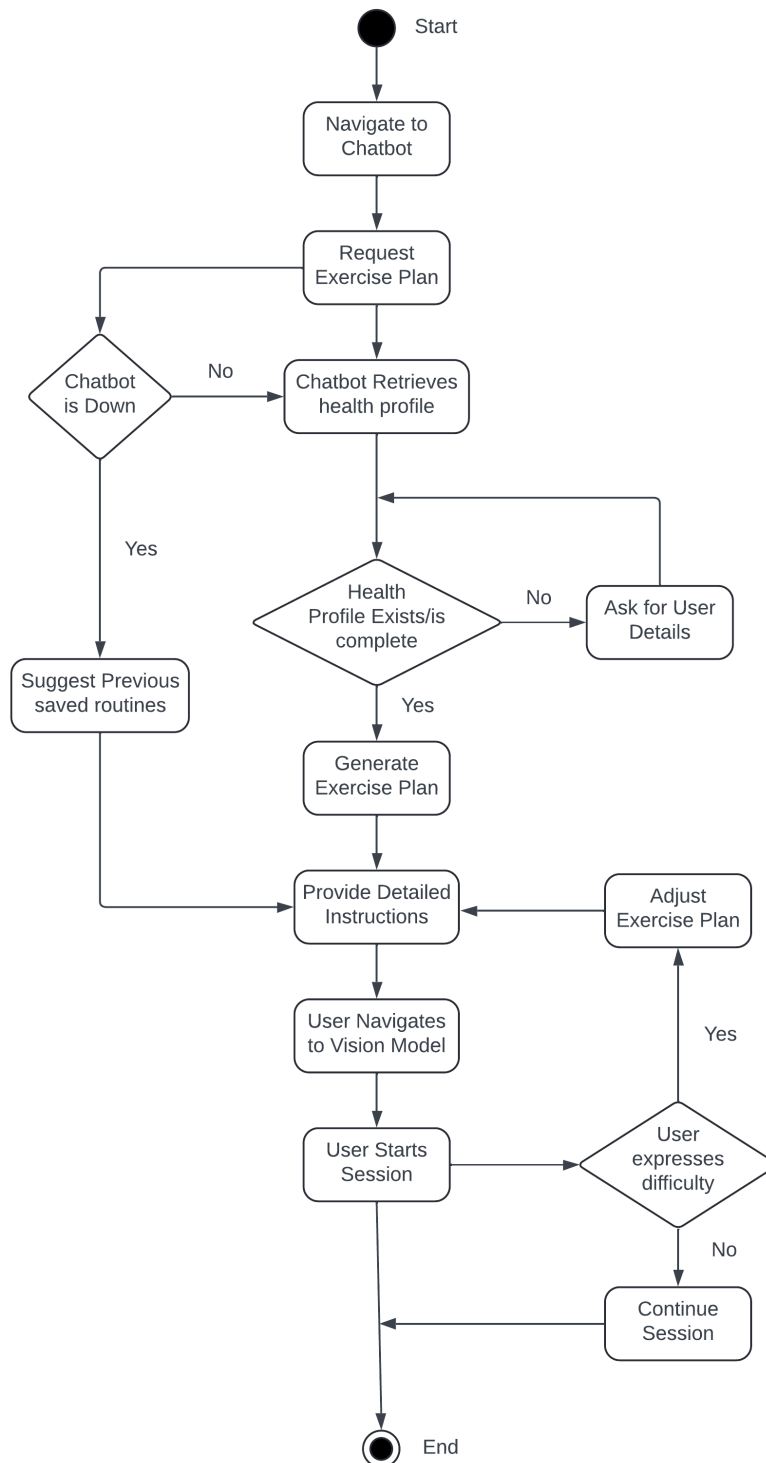
3.2.1.8 Account Management for User Profiles



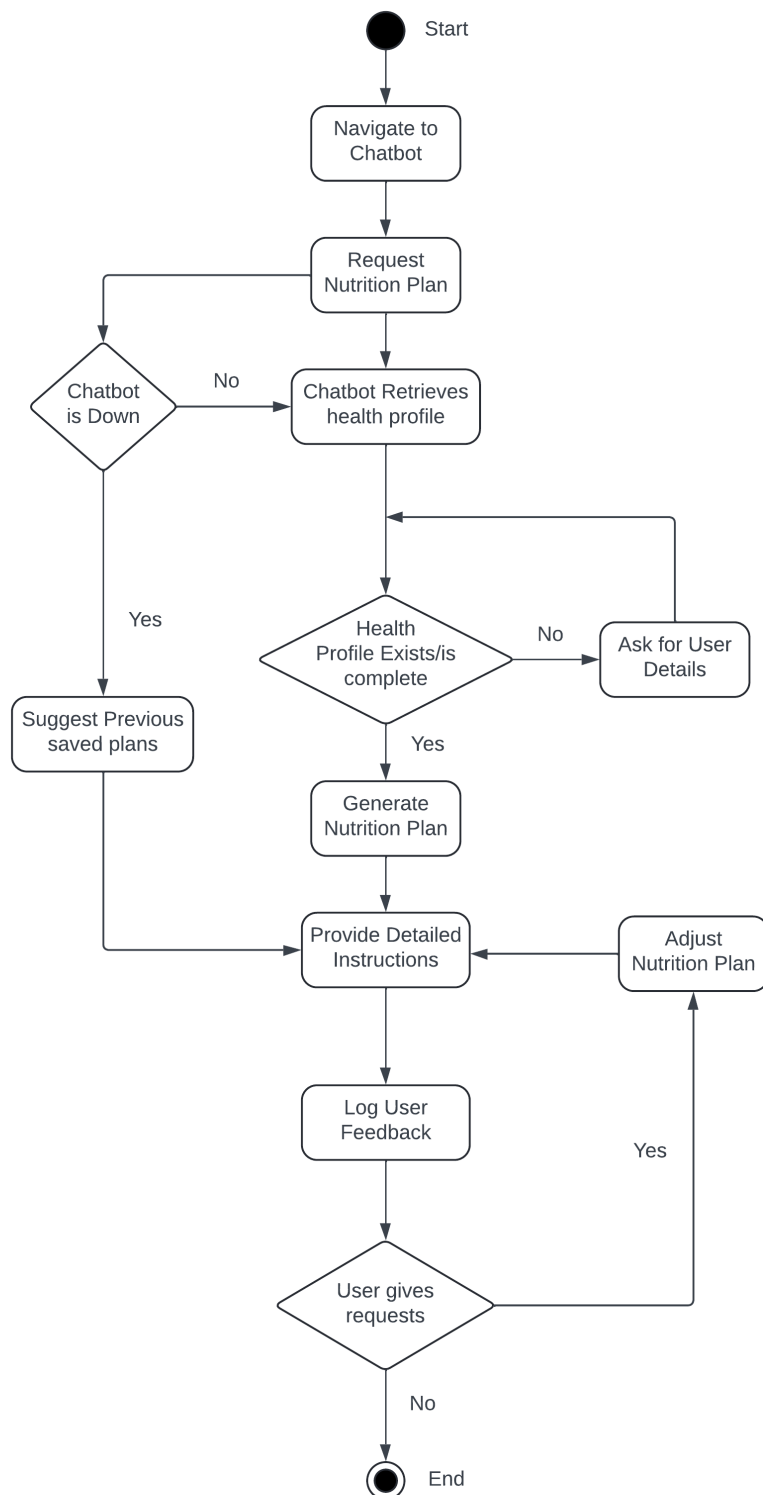
3.2.2 Activity Diagrams

The following Activity Diagrams represent the workflow of the system, illustrating the sequence of activities, decision points, and parallel processes, and how individual modules come together to create a smooth user experience.

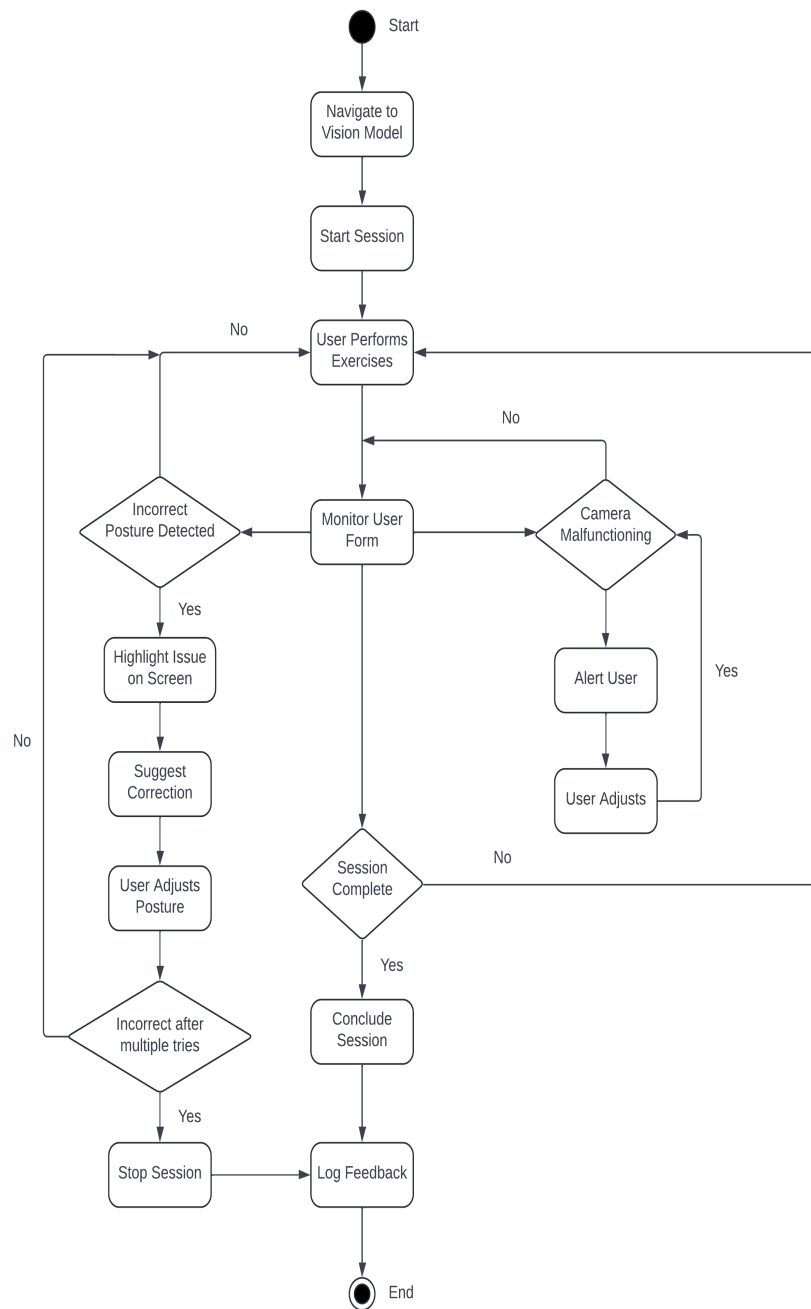
3.2.2.1 Exercise Recommendation via Chatbot



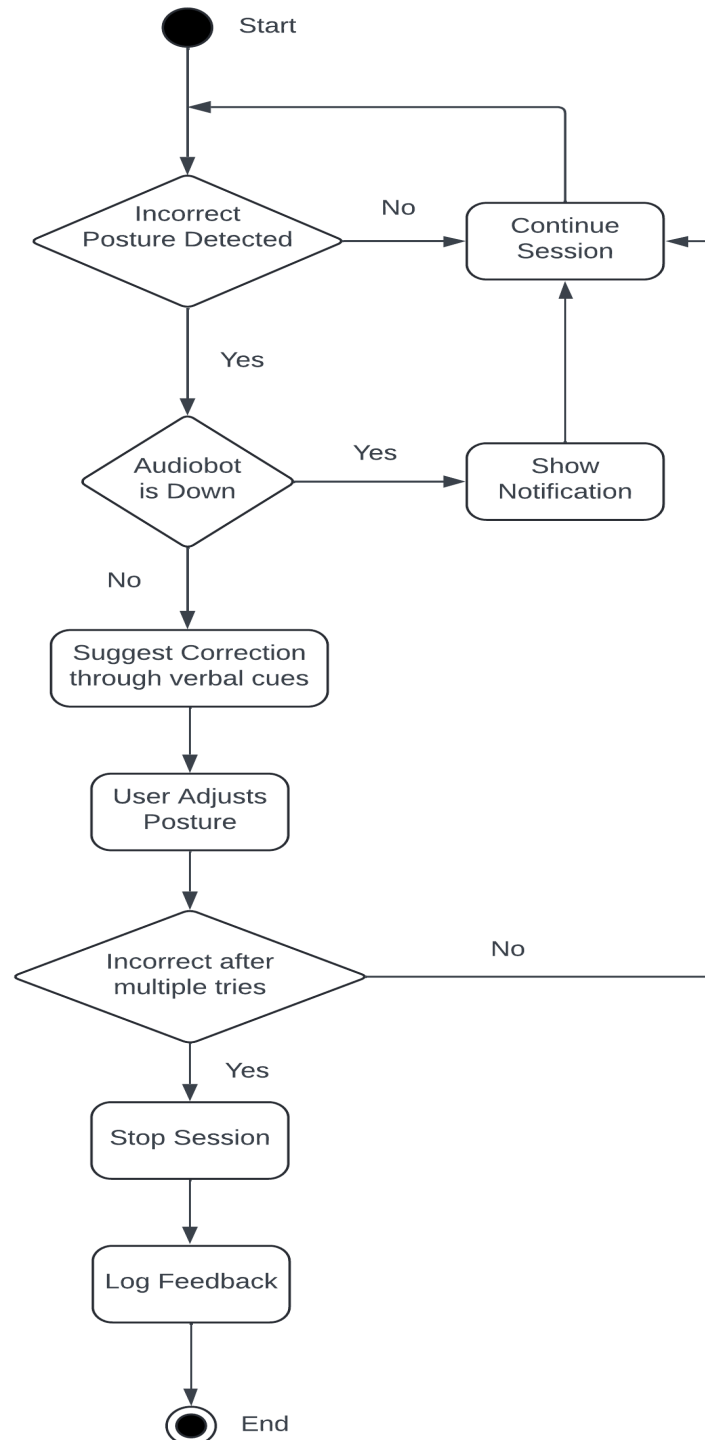
3.2.2.2 Nutrition Recommendation via Chatbot



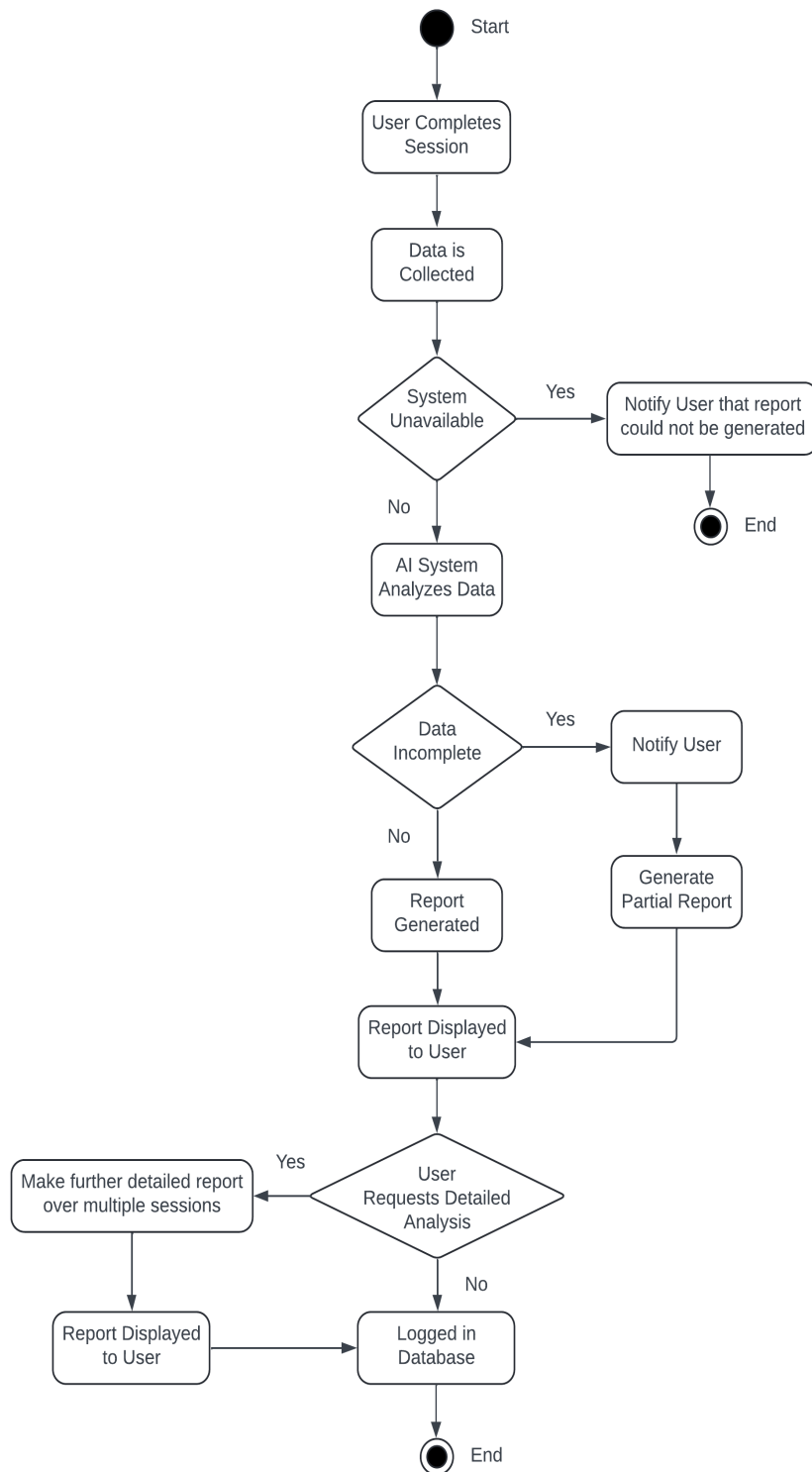
3.2.2.3 Real-Time Exercise Posture Correction



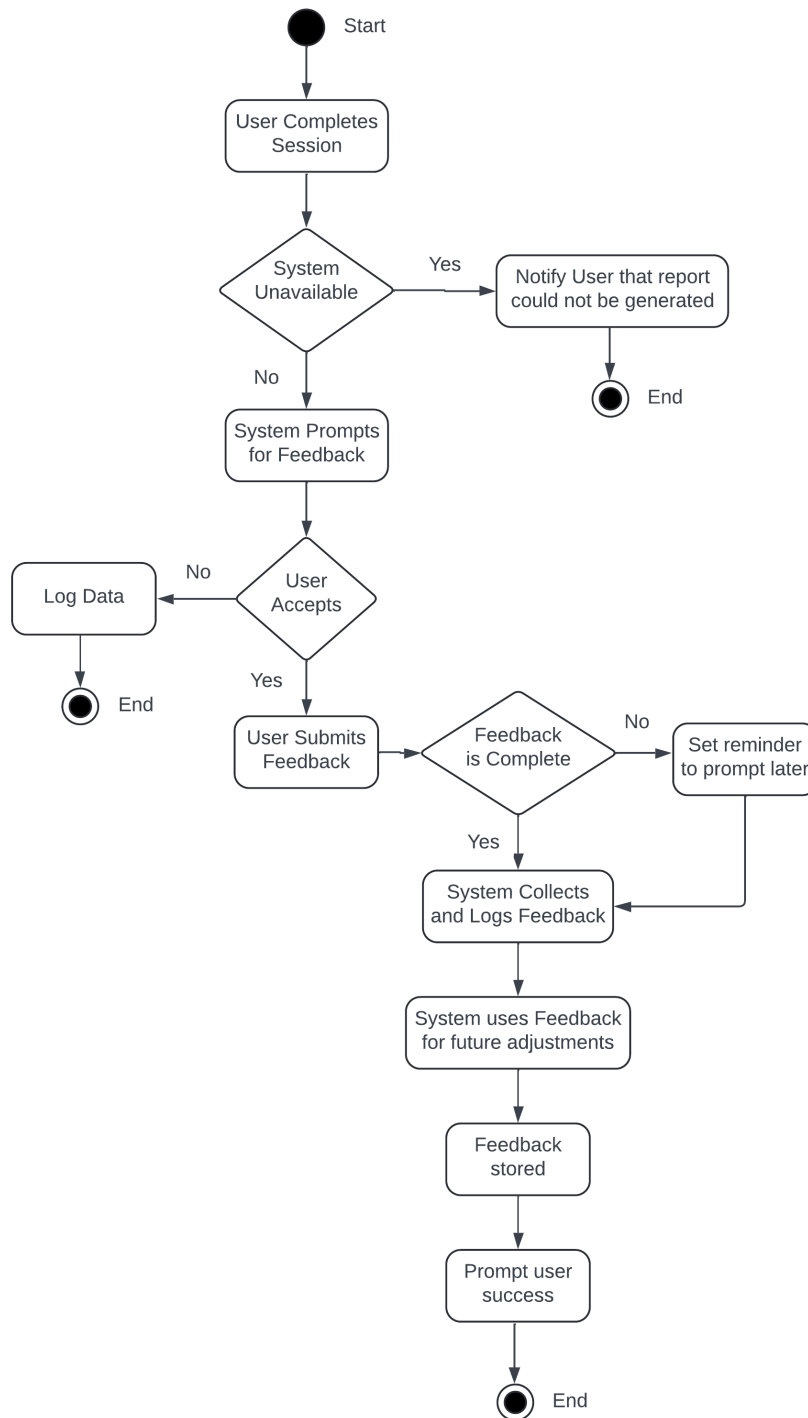
3.2.2.4 Audiobot Exercise Posture Correction



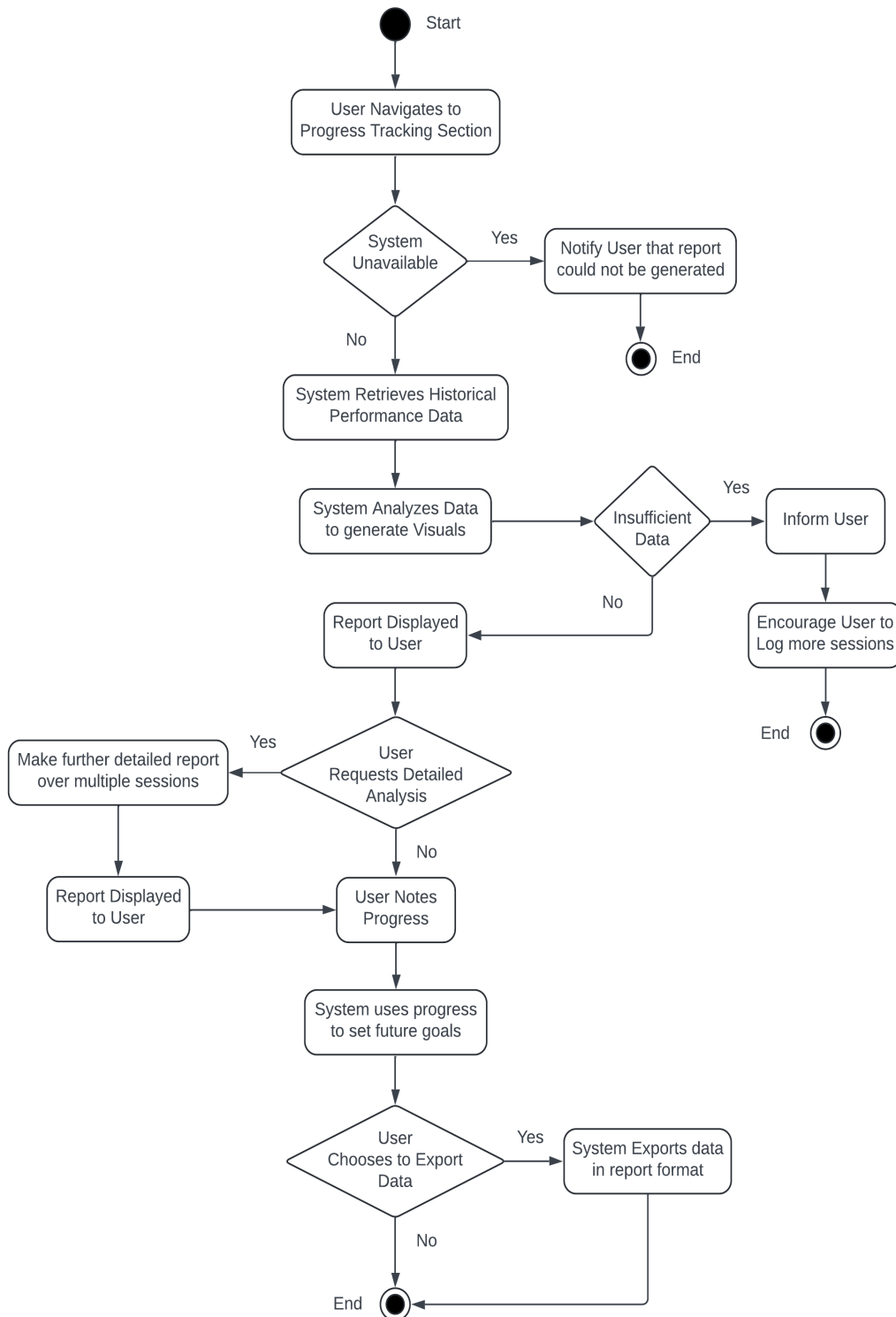
3.2.2.5 Exercise Session Report Generation



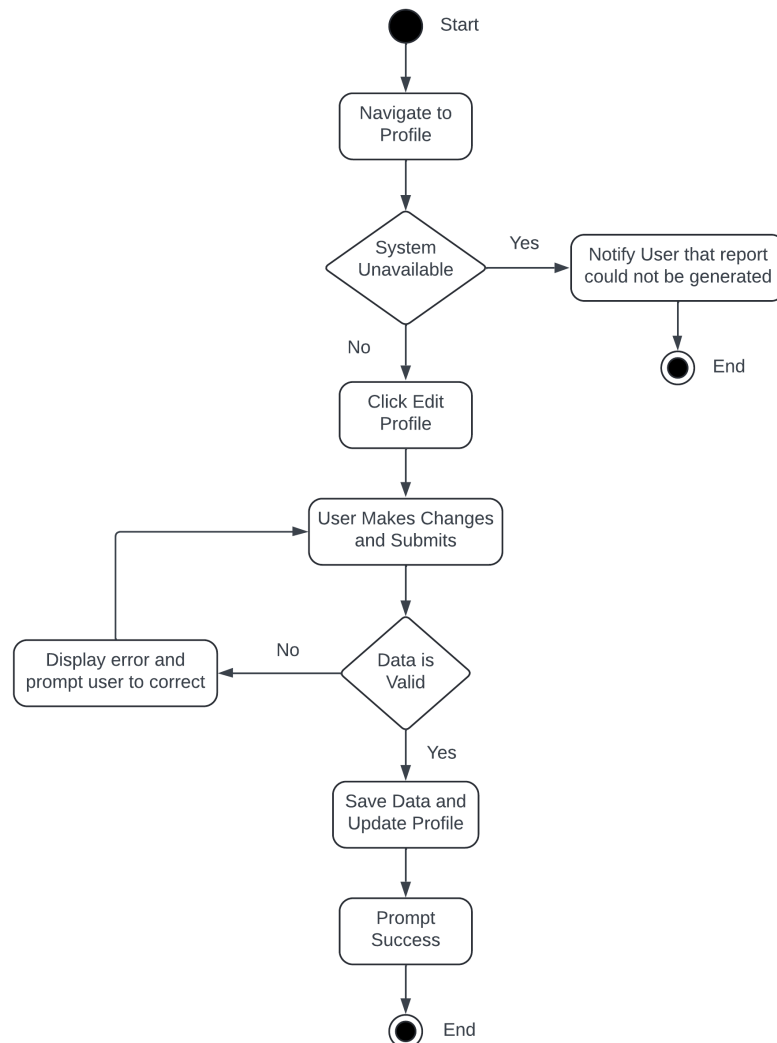
3.2.2.6 User Feedback Collection and Integration



3.2.2.7 Progress Tracking for Exercise Performance



3.2.2.8 Account Management for User Profiles



3.2.3 State Transition Diagrams

The state transition diagrams depict how the system handles different events during user interactions. They outline the various states that a user or component can enter and show how specific actions cause transitions between these states.

3.2.3.1 Starting Computer Vision

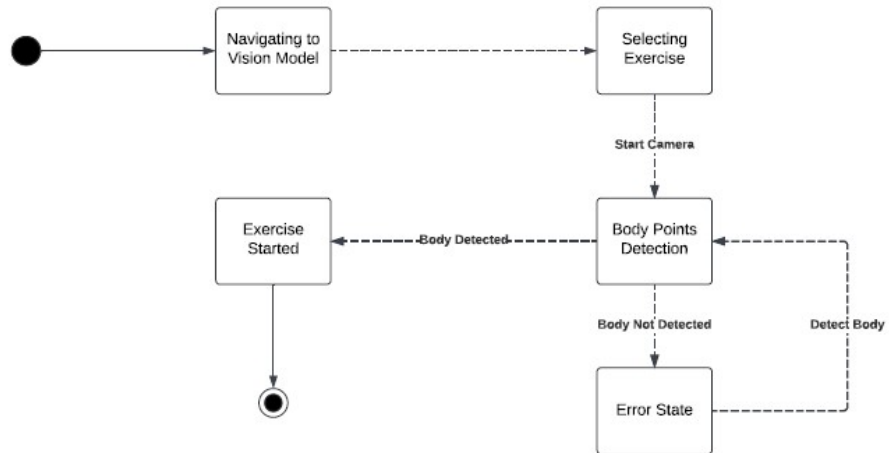


Figure 3.3: State transition diagram illustrating the initial setup of the computer vision component

3.2.3.2 Starting Chat Bot

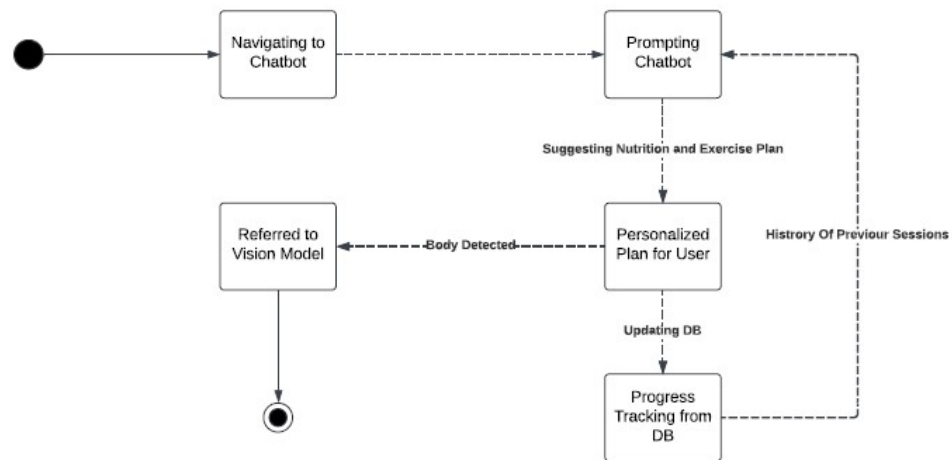


Figure 3.4: State transition diagram depicting the startup process of the chatbot module.

3.2.3.3 User Profile

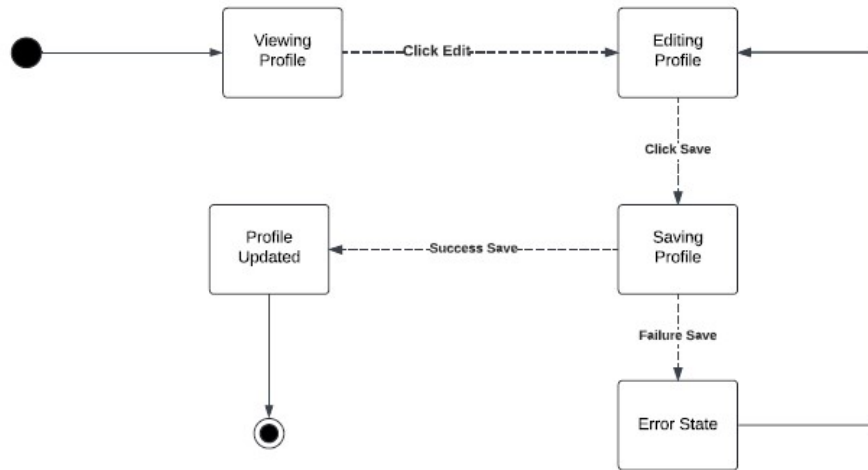


Figure 3.5: State transition diagram outlining user profile management and interactions.

3.2.3.4 End Session

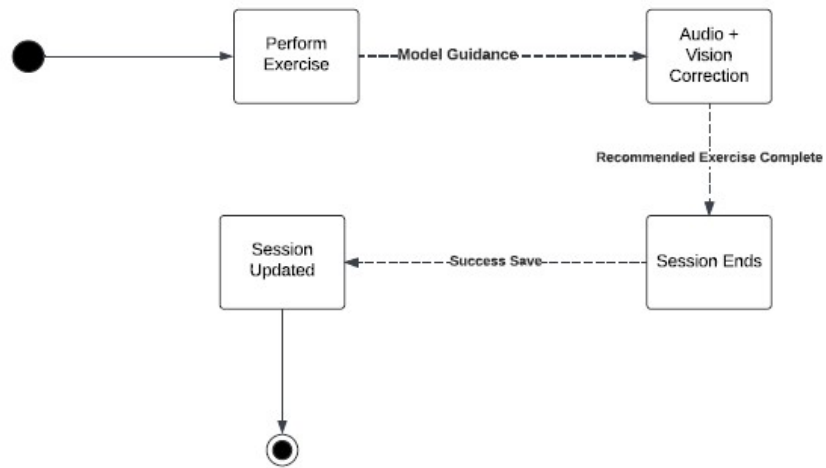


Figure 3.6: State transition diagram detailing the process of ending a user session.

3.2.4 Class Diagram

This class diagram illustrates the structure of the system by detailing the various classes, their attributes, methods, and the relationships between them. This diagram provides a clear representation of the system's object-oriented design, highlighting how different classes interact and collaborate to fulfill the desired functionalities. By mapping out these components, this diagram helps in understanding the system's organization and the roles of each class within it.

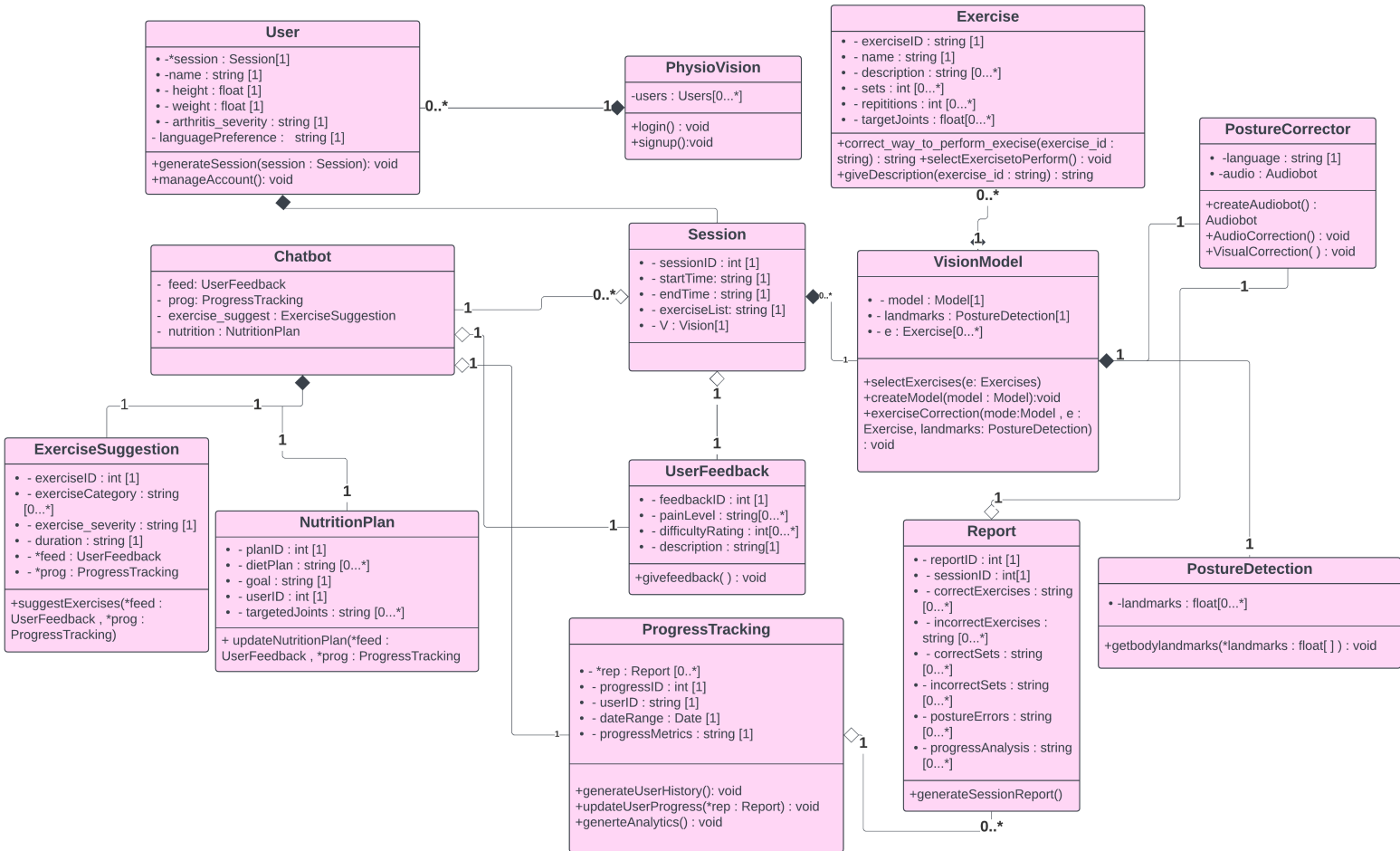


Figure 3.7: Class Diagram for PhysioVision

3.3 Data Design

This JSON schema defines key components essential for the system. The User component stores personal details and session data related to physiotherapy routines. The Session component details each routine, including timestamps and specific exercises performed. The Exercise component outlines each activity, specifying repetitions and targeted joints.

The VisionModel assists in posture correction, while UserFeedback captures pain and difficulty ratings to personalize routines. ProgressTracking monitors user recovery over time, and the Chatbot provides tailored exercise suggestions and nutrition plans. Lastly, the Report component generates performance summaries, offering insights into user progress. Together, these elements create a comprehensive solution for personalized physiotherapy management.

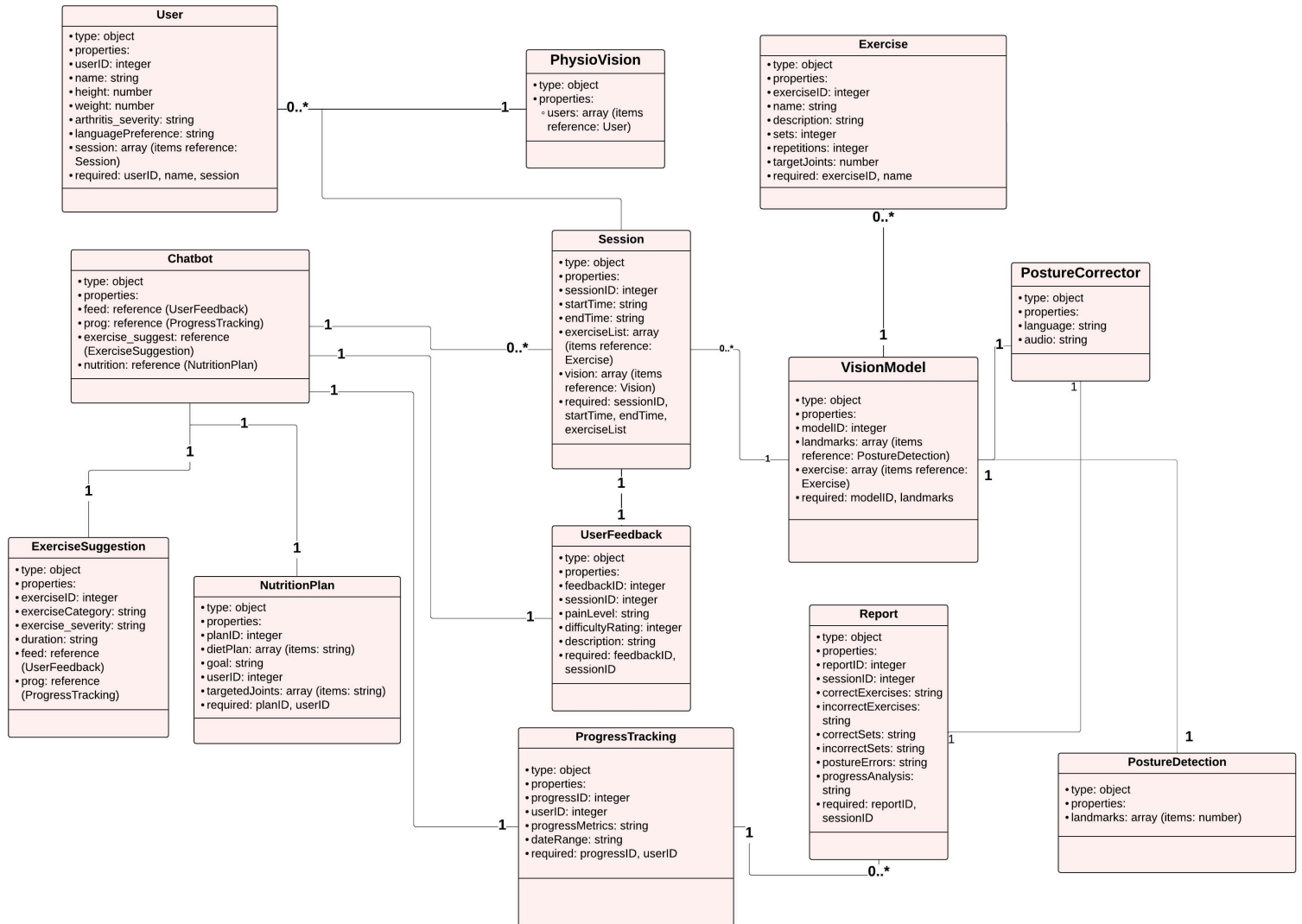


Figure 3.8: JSON SCHEMA

Bibliography

- [1] Maha Azeez, Ciara Clancy, Tom O'Dwyer, Conor Lahiff, Fiona Wilson, and Gaye Cunnane. Benefits of exercise in patients with rheumatoid arthritis: a randomized controlled trial of a patient-specific exercise programme. *Clinical Rheumatology*, 39:1783–179, 2020.
- [2] Huibin Long, Qiang Liu, Heyong Yin, Kai Wang, Naicheng Diao, Yuqing Zhang, Jianhao Lin, and Ai Guo. Prevalence trends of site-specific osteoarthritis from 1990 to 2019: Findings from the global burden of disease study 2019. *Arthritis and Rheumatology*, 74, 2022.